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Graded Assignment: Module 5 Summative Assessment Submitted
Module 5 Summary

Module 5 Summative Assessment

Your grade: 100%
Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Review Learning Objectives

Instructions

COACH
Ready to review what you've learned before starting the assignment? I'm here to help.
Quiz me on the concepts Let's chat

Assignment Details
Due Dec 22, 11:59 PM IST
Attempts 0 left (1 attempts every 8 hours)
Submitted Nov 23, 3:14 AM IST
Retry

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100%
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1. What does the term "denoise" specifically refer to in the context of diffusion models? 1 / 1 point
- Reducing the file size of images.
- Removing unwanted artifacts from compressed images.
- Reconstructing the original image from its noisy version.
- Increasing the brightness of dark images.

Correct
Correct. Like Dislike Report an issue

2. How is the forward process in Diffusion Models typically initiated? 1 / 1 point
- By sampling a clean image from the dataset.
- By applying a high-pass filter to the image.
- By compressing the image using lossless algorithms.
- By enhancing the edges in the image.

Correct
Correct.

3. In the context of diffusion models, what is the significance of the term 'KL Divergence'? 1 / 1 point
- It measures the compression efficiency of images.
- It quantifies the difference between the model's distribution and the data distribution.
- It represents the color depth of the generated images.
- It determines the resolution of the generated images.

Correct
Correct.

4. What is the primary intuition behind diffusion models? 1 / 1 point
- To directly generate images from noise.
- To iteratively add and remove noise from data.
- To classify images into different categories.
- To compress images for efficient storage.

Correct
Correct.

5. What role does the parameter t play in the training of diffusion models? 1 / 1 point
- Determines the resolution of the generated images.
- Specifies the current step in the noise addition or removal process.
- Controls the color saturation of the generated images.
- Measures the similarity between the generated and real images

Correct
Correct.

6. What is the goal of both the training and inference processes in diffusion models? 1 / 1 point
- To classify images into various categories with high accuracy.
- To generate high-resolution images from low-resolution inputs.
- To generate images that are indistinguishable from real images.
- To compress images without losing quality.

Correct
Correct.

7. What is the 'Mean of Gaussian' used for in the context of diffusion models? 1 / 1 point
- To define the initial distribution of images before training.
- As a parameter for adjusting the brightness of the generated images.
- In the denoising step, to predict the clean image from a noisy observation.
- To categorize generated images into different classes based on their content.

Correct
Correct.

8. What does the term 'conditioning' in diffusion models refer to? 1 / 1 point
- Adjusting the brightness and contrast of images.
- The process of training models exclusively on grayscale images
- Incorporating specific information to guide the generation process.
- Reducing the size of the model for faster inference.

Correct
Correct.

9. During the training of a diffusion model, what is the role of the noise predictor? 1 / 1 point
- To predict the clean image directly from the noise.
- To estimate the amount of noise to add at each step.
- To predict the original noise added to the clean image.
- To classify the type of noise added to the image.

Correct
Correct.

10. How do diffusion models differ from Variational Autoencoders (VAEs)? 1 / 1 point
- Diffusion models do not use an encoder-decoder architecture.
- Diffusion models only encode data but do not decode it.
- VAEs add noise to data, while diffusion models remove noise.
- VAEs use a fixed noise level, while diffusion models use a variable one.

Correct
Correct.

11. Which of the following best describes the forward process in diffusion models? 1 / 1 point
- Removing noise from an image.
- Directly generating clean images from noise.
- Gradually adding noise to an image.
- Compressing images for storage

Correct
Correct.

12. In diffusion models, how is the variance of the noise adjusted during training? 1 / 1 point
- It is fixed throughout the process.
- It increases linearly with each step.
- It is tunable based on the training dynamics.
- It is inversely proportional to the image size.

Correct
Correct.

13. Why is maximizing likelihood equivalent to minimizing KL divergence in the context of diffusion models? 1 / 1 point
- Because it leads to faster training of the models.
- It ensures the generated images are of the highest possible resolution.
- Minimizing KL divergence aligns the model's generated data distribution closely with the real data distribution
- Because it enhances the color accuracy of the generated images.

Correct
Correct.

14. Which of the following best describes the concept of 'Denoising Diffusion Probabilistic Models' (DDPM)? 1 / 1 point
- A model that predicts future stock prices based on past data.
- A framework for compressing images without significant loss of quality.
- A generative model that learns to reverse a diffusion process to generate data.
- A method for removing background noise from audio recordings.

Correct
Correct.

15. What is a critical step in the inference process of diffusion models? 1 / 1 point
- Increasing the resolution of the input image.
- Classifying the image into predefined categories.
- Sampling a noise distribution to start the reverse process.
- Directly generating the final image without noise.

Correct
Correct.